CHEAT SHEET

Logreturn: $y \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right) t\ ;\ \sigma\sqrt{t}\right)$

Price :

$$S_T = S_0 e^y$$
$$E(S_T) = S_0 e^{\mu}$$
$$\sigma(S_T) = S_0 e^{\mu}\sqrt{e^{\sigma^2} - 1}$$

GBM: $dS = \mu S dt + \sigma S dw$

$$dY = (\mu - \frac{\sigma^2}{2})dt + \sigma dw$$

$$\Delta Y = (\mu - \frac{\sigma^2}{2})\Delta t + \sigma \Delta w$$

$$N\left(\frac{K-\mu}{\sigma}\right) = 1 - \alpha$$

$$N^{-1}(1-\alpha) = \frac{K-\mu}{\sigma}$$

$$K = \mu + N^{-1}(1-\alpha) * \sigma$$

$$N^{-1}(0.01) = -2.326$$
$$N^{-1}(0.05) = -1.645$$
$$N^{-1}(0.95) = 1.645$$
$$N^{-1}(0.99) = 2.326$$

$$ES = \mu - \sigma \frac{e^{-(N^{-1}(\alpha))^2/2}}{\sqrt{2\pi}(1-\alpha)}.$$

$$D* = \frac{\Delta P / P}{\Delta r}$$

$$\frac{\Delta P}{P} = D^* \cdot \Delta r$$

$$\sigma_{Bond} = \sigma\left(\frac{\Delta P}{P}\right) = (-D^*) \cdot \sigma(\Delta r)$$

$$VaR_{Bond} = multiplier \cdot \sigma_{Bond} = multiplier * (-D^*) \cdot \sigma(\Delta r)$$

$$c = SN(d_1) - PXN(d_2)$$

$$d_1 = \frac{\ln(\frac{S_0}{X}) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(\frac{S_0}{X}) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$u_1, u_2 \sim N(0,1)$$

$x_1 = u_1$ és $x_2 = u_1 \cdot \rho + u_2 \cdot \sqrt{1-\rho^2}$

$$WCDR_i = N\left[\frac{N^{-1}(PD_i) + \sqrt{\rho}N^{-1}(0{,}999)}{\sqrt{1-\rho}}\right]$$

$$EL = \sum_i EAD_i \cdot LGD_i \cdot PD_i$$

$$UL = \sum_i EAD_i \cdot LGD_i . (WCDR_i - PD_i)$$

$$E = A \cdot N(d_1) - P \cdot D \cdot N(d_2)$$
$$\sigma_E \cdot E = \frac{\partial E}{\partial A}\sigma_A \cdot A$$

$$DD = \frac{\ln(A_0 / D) + (\mu - \sigma_A^2 / 2) \cdot T}{\sigma_A \cdot \sqrt{T}}$$

EDF=1-N(DD)

$$CVA = \sum_{i=1}^{n} (1-R) q_i v_i$$